

Universal shaper

3.5 Quick return mechanism

The ram moves at a comparatively slower speed during the forward cutting stroke. During the return stroke, the mechanism is so designed to make the tool move at a faster rate to reduce the idle return time. This mechanism is known as quick return mechanism.

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At the ram moves at a faster rate during stroke, the time taken becomes less. The total machining time decreases and the rate of production increases. The following mechanisms are used for quick return of the ram.

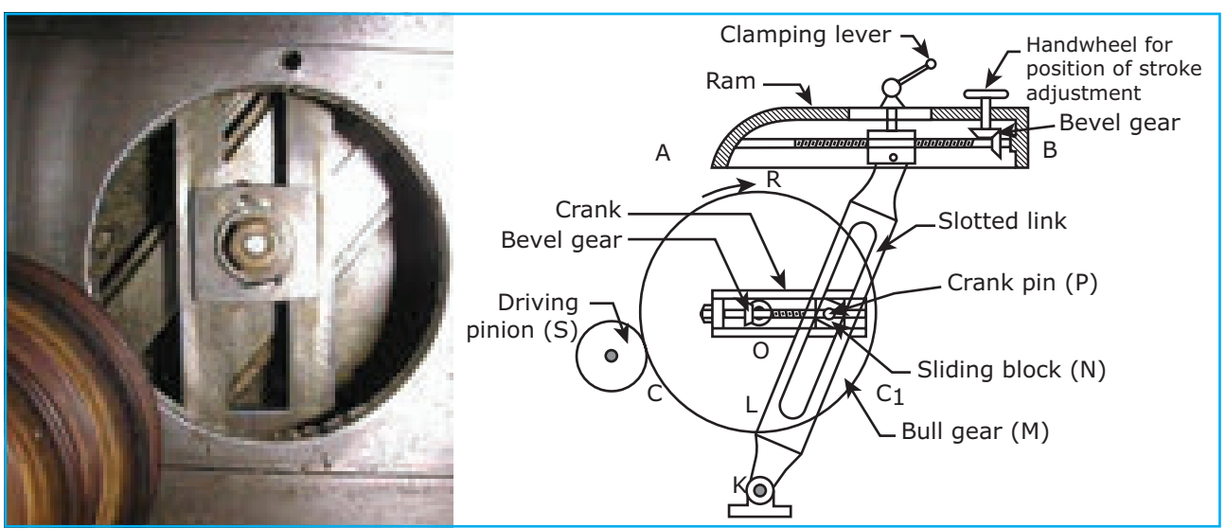
1. Crank and slotted link mechanism
2. Hydraulic mechanism
3. Whitworth mechanism

Crank and slotted link mechanism

An electrical motor runs the driving pinion(S) at a uniform speed. This pinion makes the bull gear (M) to rotate at a uniform speed. Bull gear is a large gear fitted inside the column. The point 'O' is the centre of the bull gear. A slotted link having a long slot along its length is pivoted about the point 'K'. A sliding block 'N' is fitted inside the slot and slides along the length of the slotted link. 'P' is the crank pin and 'OP' can be considered as a crank.

Figure shows the crank & slotted link mechanism.

When the bull gear rotates, the sliding block also rotates in the crank pin circle. This arrangement provides a rocking movement to the rocker arm. As



Crank and Slotted link Mechanism

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